

REVIEW
Diabetic Complications in Cardiovascular Health

Recent advances associated with cardiometabolic remodeling in diabetes-induced heart failure

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Abstract

Diabetes mellitus (DM) is characterized by chronic hyperglycemia, and despite intensive glycemic control, the risk of heart failure in patients with diabetes remains high. Diabetes-induced heart failure (DHF) presents a unique metabolic challenge, driven by significant alterations in cardiac substrate metabolism, including increased reliance on fatty acid oxidation, reduced glucose utilization, and impaired mitochondrial function. These metabolic alterations lead to oxidative stress, lipotoxicity, and energy deficits, contributing to the progression of heart failure. Emerging research has identified novel mechanisms involved in the metabolic remodeling of diabetic hearts, such as autophagy dysregulation, epigenetic modifications, polyamine regulation, and branched-chain amino acid (BCAA) metabolism. These processes exacerbate mitochondrial dysfunction and metabolic inflexibility, further impairing cardiac function. Therapeutic interventions targeting these pathways—such as enhancing glucose oxidation, modulating fatty acid metabolism, and optimizing ketone body utilization—show promise in restoring metabolic homeostasis and improving cardiac outcomes. This review explores the key molecular mechanisms driving metabolic remodeling in diabetic hearts, highlights advanced methodologies, and presents the latest therapeutic strategies for mitigating the progression of DHF. Understanding these emerging pathways offers new opportunities to develop targeted therapies that address the root metabolic causes of heart failure in diabetes.

cardiac metabolism; diabetes mellitus; heart failure mechanisms; metabolic therapies; mitochondrial dysfunction

INTRODUCTION

Diabetes mellitus (DM) is a global public health crisis, currently affecting over 537 million adults worldwide, with this number expected to rise significantly in the coming years (1). In 2021, 11.6% of the US population had DM, contributing to a substantial economic burden, with the total cost of diagnosed diabetes in the United States reaching \$412.9 billion in 2022 (2). Although intensive glycemic control has successfully mitigated microvascular complications, it has proven insufficient in reducing the risk of heart failure in patients with diabetes, highlighting the need for targeted strategies that address the underlying metabolic disturbances contributing to heart failure (3). Recent studies highlight the promising potential of metabolic remodeling as a therapeutic approach to managing diabetes-induced heart failure (DHF). This approach addresses the

underlying metabolic disturbances, offering a new avenue to improve cardiac function and outcomes in patients with diabetes (4). This review explores the key molecular pathways and signaling networks underlying diabetes-induced cardiac metabolic dysfunction. It highlights advanced technologies and imaging tools used to study these processes. A deeper understanding of these metabolic mechanisms is essential for developing effective interventions to address the growing burden of heart failure in diabetes.

**CELLULAR, PATHOLOGICAL, AND CLINICAL
MANIFESTATIONS OF DIABETES-INDUCED
HEART FAILURE**

DM is a chronic metabolic disorder characterized by elevated blood sugar levels (hyperglycemia) due to the body's

inability to produce enough insulin or effectively use the insulin it produces. The two major types of DM are type 1 DM (T1DM) and type 2 DM (T2DM). T1DM results from the immune system attacking insulin-producing cells in the pancreas, leading to insulin deficiency whereas T2DM is caused by insulin resistance, where cells fail to respond properly to insulin, often coupled with a progressive decline in insulin production. Diabetes leads to various cardiac complications, including the buildup of fatty deposits in arteries, increasing the risk of atherosclerosis (5–7). It also causes microvascular issues, reducing blood flow and oxygen delivery to the heart (8–11) (Fig. 1). In addition, diabetes directly impairs cardiac muscle function, leading to diabetic cardiomyopathy (DMCM), which raises the risk of heart failure (12–14).

Clinical manifestations of DHF can include a wide range of symptoms that stem from both structural and functional impairments in the heart. Early signs of heart failure in patients with diabetes often involve left ventricular diastolic dysfunction, which leads to symptoms like exertional dyspnea (shortness of breath during activity) and fatigue (15–17). As heart failure progresses, more severe symptoms like peripheral edema, pulmonary congestion, orthopnea (difficulty breathing while lying down), and paroxysmal

nocturnal dyspnea (sudden night-time breathlessness) emerge (18, 19). In addition, individuals with DHF are more likely to experience arrhythmias due to structural and electrical remodeling of the heart tissue, exacerbated by the presence of hyperglycemia and insulin resistance (20). In advanced stages, patients may develop cardiomegaly and exhibit elevated biomarkers, such as brain natriuretic peptide (BNP) or N-terminal pro-BNP (NT-proBNP), indicative of increased cardiac stress and heart failure progression. Furthermore, the presence of microvascular dysfunction and myocardial fibrosis, often seen in DMCM, can further impair cardiac output and exacerbate heart failure symptoms, making management more complex (18, 21). Diabetes increases the clinical burden when overlapped with other comorbidities like hypertension and coronary artery disease. It complicates the presentation of the disease requiring a comprehensive evaluation to differentiate between heart failure of diabetic origin and other etiologies.

MULTIFACTORIAL MECHANISMS AFFECTING DIABETES-INDUCED HEART FAILURE

Heart failure in individuals with diabetes exhibits distinct clinical features, where increased intramyocardial lipids are

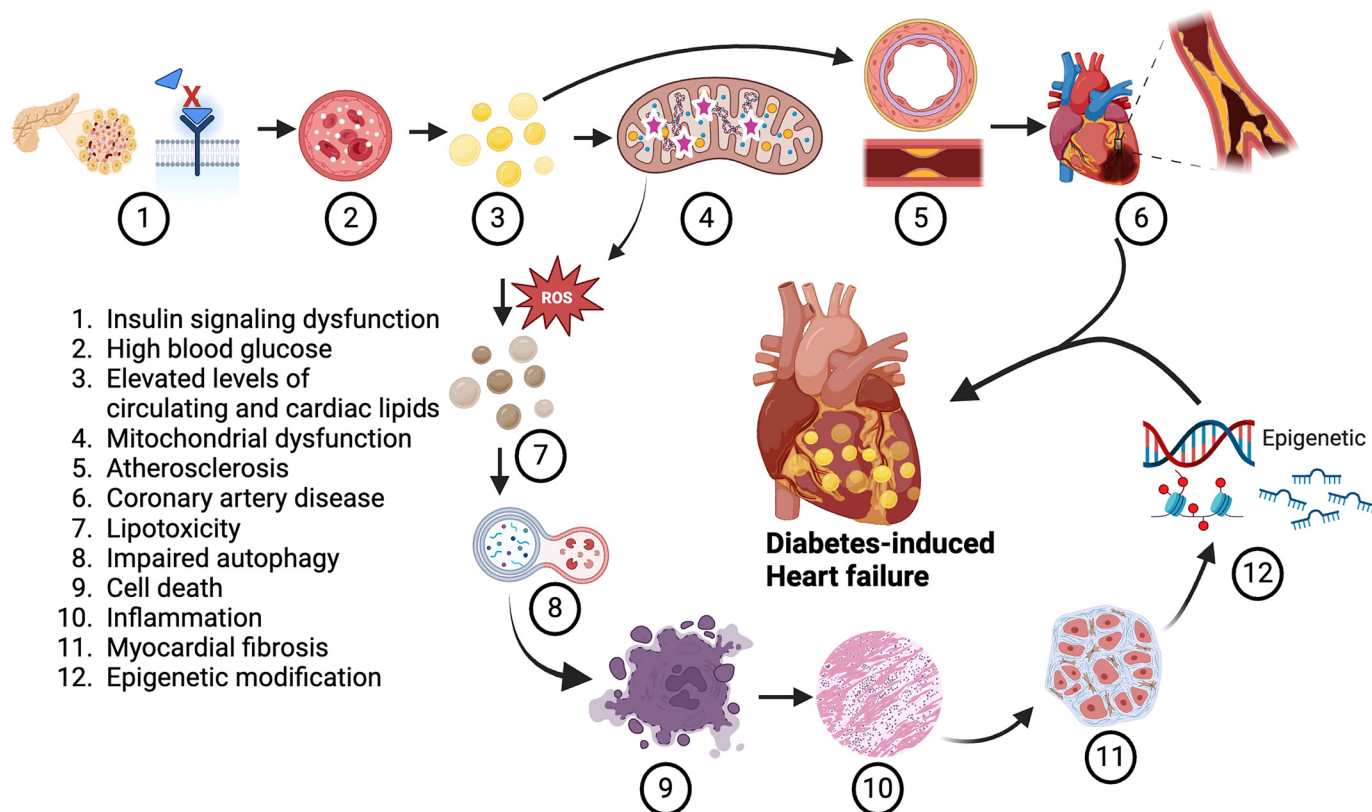


Figure 1. Mechanisms of diabetes-induced heart failure (DHF). This schematic representation illustrates the potential causes of heart failure in diabetes. Obesity and pancreatic β -cell destruction are critical contributors to the development of diabetes. Elevated glucose levels in the bloodstream increase blood viscosity and glycosylated hemoglobin levels. Diabetes also induces mitochondrial dysfunction, leading to elevated production of reactive oxygen species (ROS), due to a metabolic remodeling in the heart. In nondiabetic hearts, glucose and fatty acids are used in a balanced manner for ATP production. However, in diabetic hearts, glucose uptake and oxidation are significantly reduced due to insulin inactivity and the downregulation of glucose transporters such as GLUT4. Meanwhile, there is an increased reliance on fatty acids, with heightened uptake facilitated by transporters such as CD36 and fatty acid transport proteins (FATP), resulting in excessive fatty acid oxidation. This metabolic remodeling leads to the accumulation of toxic lipid intermediates, such as diacylglycerol (DAG) and ceramides, contributing to lipotoxicity, mitochondrial dysfunction, and impaired cardiac efficiency. These metabolic remodeling events alter regulatory molecules, including epigenetic modifiers, maintaining homeostatic gene levels in cells. These changes foster oxidative stress and pathological remodeling, ultimately leading to heart failure. Images were created with a licensed version of BioRender.com.

linked to cardiac dysfunction (22, 23). Women with diabetes are more likely to develop heart failure with preserved ejection fraction (HFpEF), whereas men more commonly experience heart failure with reduced ejection fraction (HFrEF) (24, 25) (Fig. 2). The alterations in metabolites and their signaling pathways during HFpEF have been detailed in a recent review article (26). Metabolic syndrome (MetS) or cardiometabolic syndrome (CMetS), a cluster of metabolic disorders including insulin resistance, dyslipidemia, and hypertension, often coexists with diabetes and compounds the risk of developing heart failure (27, 28). The overlap in metabolic disturbances between MetS and DMCM suggests that insulin resistance and increased fatty acid utilization may be common drivers of cardiac dysfunction (29, 30). Mitochondrial dysfunction and oxidative stress are critical mediators of metabolic remodeling in diabetic hearts, contributing to heart failure with preserved ejection fraction (HFpEF) in this population (31, 32). A hallmark of DHF is the remodeling of cardiac metabolism, which is closely linked to mitochondrial dysfunction, increased oxidative stress, and impaired energy production (33). These cardiometabolic disturbances include insulin resistance, reduced glucose utilization, increased fatty acid oxidation, and lipotoxicity. However, the precise molecular mechanisms driving these changes in diabetic hearts remain incompletely understood. In diabetic hearts, insulin resistance causes a marked change in cardiac metabolism where fatty acid substrates are more used than glucose to produce energy (34). The metabolic flexibility typically seen in healthy cardiomyocytes is lost, resulting in over-reliance on fatty acids for ATP generation (35). This alternative substrate utilization not only increases oxygen demand but also promotes lipotoxicity and mitochondrial dysfunction through the accumulation of lipid intermediates such as ceramides and diacylglycerol

(36–38). This imbalance leads to mitochondrial dysfunction, characterized by the production of excessive reactive oxygen species (ROS) and oxidative stress, which further impairs energy production and exacerbates contractile dysfunction, ultimately contributing to heart failure (34).

Key Metabolic Changes in Diabetes-Induced Heart Failure

DHF is characterized by profound metabolic remodeling, which impairs cardiac function and contributes to disease progression (39). The interplay between insulin resistance, altered substrate utilization, mitochondrial dysfunction, and oxidative stress leads to significant disruptions in the heart's ability to produce energy efficiently. In the following sections, we explore the key metabolic alterations in DHF, focusing on their molecular mechanisms and clinical relevance.

Insulin resistance and glucose utilization deficiency.

Insulin resistance, a fundamental feature of DMCM, impairs glucose uptake and utilization in cardiomyocytes (40). This reduction in glucose oxidation forces the heart to rely more heavily on fatty acid oxidation for energy (41–44). Insulin signaling dysregulation via PI3K/Akt pathways also impairs mitochondrial function and contributes to oxidative stress (45–47).

Increased fatty acid oxidation and lipotoxicity.

In diabetes, reduced glucose availability leads to an increased reliance on fatty acid oxidation (FAO) as a primary energy source (41, 48). Elevated circulating free fatty acids (FFAs) and enhanced expression of fatty acid transport proteins (e.g., CD36 and FATP) drive this metabolic reprogramming (49). However, FAO is less oxygen-efficient compared with

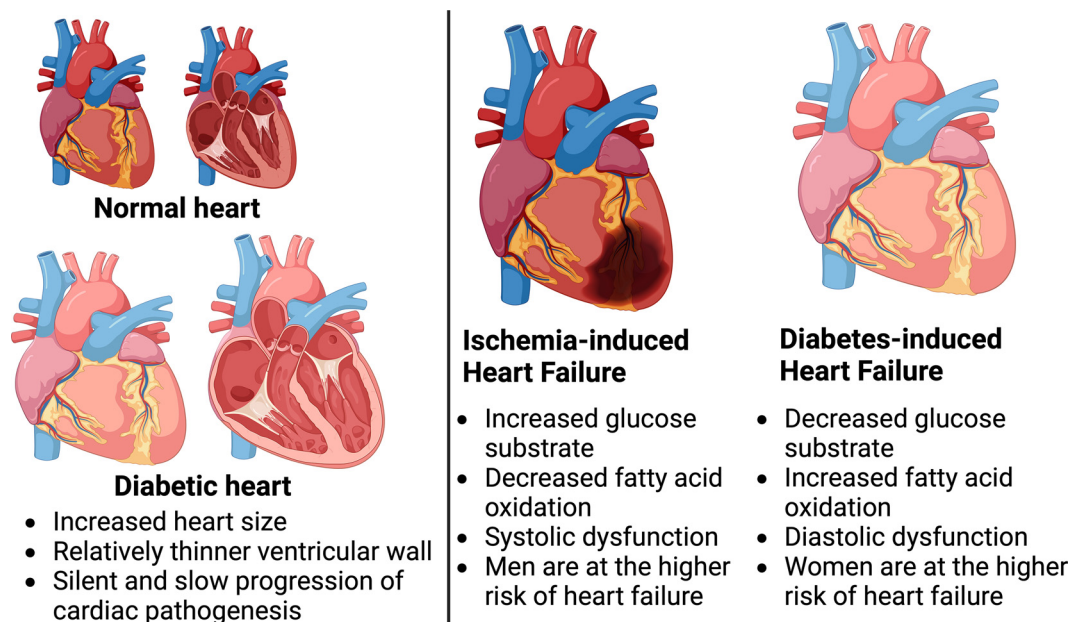


Figure 2. Hallmarks of diabetes-induced heart failure (DHF). DHF is characterized by several distinctive features. The condition involves a reduced availability of glucose metabolic substrates, which results in mitochondrial dysfunction. This leads to increased reliance on fatty acid oxidation in the diabetic heart. Notably, females have a higher risk of developing heart failure compared with males. Moreover, the progression of diabetic heart failure typically advances from diastolic dysfunction to systolic dysfunction, contrasting with ischemic heart failure, where systolic dysfunction precedes heart failure. Images were created with a licensed version of BioRender.com.

glucose oxidation. This “metabolic inefficiency” increases oxygen demand and contributes to myocardial hypoxia (50). It occurs when cells or tissues consume more energy (ATP) to perform the same functions or produce less energy from the same amount of fuel, often due to impaired or dysfunctional metabolic pathways. This leads to suboptimal energy production and greater energy expenditure (51). Additionally, increased FAO promotes lipotoxicity, where excess fatty acids accumulate as toxic lipid intermediates, such as diacylglycerol (DAG) and ceramides, within cardiomyocytes (52). These lipids activate proinflammatory and proapoptotic pathways, including protein kinase C (PKC) and c-Jun N-terminal kinase (JNK), exacerbating insulin resistance, oxidative stress, and cardiomyocyte death (53). This accumulation of toxic lipids further impairs cardiac function and contributes to the structural remodeling of the heart, as observed in DMCM (54, 55).

Mitochondrial dysfunction and energetic deficits.

Mitochondrial dysfunction includes disrupted fission/fusion, impaired electron transport chain activity, and reduced biogenesis (56–58). Increased ROS production exacerbates oxidative damage, impairing ATP production and promoting cardiomyocyte cell death (59, 60). One of the key issues is the elevated production of ROS. Hyperglycemia and fatty acid overload, a characteristic feature of diabetes, lead to an excessive accumulation of ROS in the mitochondria (61). These highly reactive molecules cause oxidative damage to mitochondrial DNA (mtDNA), proteins, and lipids (62). This oxidative stress disrupts mitochondrial functions, such as oxidative phosphorylation (OXPHOS), reducing ATP production efficiency and further damaging the electron transport chain (ETC) (63). This creates a vicious cycle in which the impaired ETC generates even more ROS, further exacerbating mitochondrial damage. Additionally, diabetes induces structural changes within the mitochondria (64, 65). The heart contains two main types of mitochondria: interfibrillar mitochondria (IFM), located between muscle fibers and essential for sustaining cardiac contractility, and subsarcolemmal mitochondria (SSM), situated beneath the plasma membrane (66). In patients with T2DM, analysis of mitochondrial subpopulations from atrial appendages reveals a significant decrease in SSM respiration, particularly through Complexes I and IV. In contrast, no changes were observed in IFM respiration compared with nondiabetic individuals (64). In the hearts of T1DM mice, the IFM subpopulation exhibits impaired activity due to a reduction in cardiolipin synthase protein content (67, 68). Another crucial aspect of mitochondrial dysfunction in diabetes is the dysregulation of mitophagy, the process by which cells eliminate damaged mitochondria (69). In healthy cells, mitophagy ensures that damaged or dysfunctional mitochondria are degraded and replaced by new, functioning mitochondria. In diabetes, mitophagy is impaired, leading to the accumulation of damaged mitochondria that continue to produce excessive ROS and fail to maintain proper membrane potential (70). This persistent damage triggers mitochondrial depolarization, further inhibiting ATP production and driving the heart toward apoptosis, or programmed cell death. The accumulation of dysfunctional mitochondria weakens the cardiac muscle and promotes the development of heart failure

(70, 71). In addition, mitochondrial biogenesis, the process responsible for creating new mitochondria, is also compromised in diabetic hearts of humans and mice (72, 73). Key regulators of this process, such as peroxisome proliferator-activated receptor- γ coactivator 1- α (PGC-1 α), are downregulated in diabetic hearts, leading to a reduced capacity for mitochondrial renewal (73). This decline in mitochondrial biogenesis further worsens mitochondrial health, creating an environment in which the heart cannot efficiently adapt to increased metabolic demands. The spatial distribution of mitochondria within cardiac cells is altered in diabetes (68). Proper mitochondrial positioning within the cell is essential for effective energy distribution. In diabetes, this spatial organization is disrupted, impairing the delivery of ATP to areas of the cell where it is most needed, such as the contractile machinery. This spatial disorganization contributes to the overall decline in cardiac function observed in DMCM (74). These multifaceted changes in mitochondrial dynamics, ROS production, structural integrity, mitophagy, biogenesis, and spatial organization collectively contribute to the progression of heart failure in diabetes. Addressing these mitochondrial abnormalities could provide therapeutic targets to mitigate diabetic heart disease and improve cardiac function in affected patients (69, 75).

Oxidative stress and ROS overproduction.

Oxidative stress plays a critical role in the progression of DHF. In diabetic hearts, elevated levels of ROS, produced primarily by mitochondria and NADPH oxidases, contribute to cellular damage (76). Excess ROS leads to lipid peroxidation, DNA damage, and the inactivation of key metabolic enzymes involved in ATP production (77). The oxidative damage to mitochondria further impairs oxidative phosphorylation, creating a vicious cycle of energy deficits and further ROS generation (78). Furthermore, antioxidant defenses in diabetic hearts are often compromised, as indicated by reduced levels of essential antioxidants such as superoxide dismutase (SOD) and glutathione (GSH) (79, 80). This imbalance between ROS production and antioxidant capacity accelerates cardiac injury, promoting inflammation, fibrosis, and myocardial cell death (81). Therapeutic strategies targeting oxidative stress may hold promise in reducing cardiac dysfunction in diabetes.

Cell death and fibrosis.

Metabolic stress and mitochondrial dysfunction promote cardiomyocyte death via at least six distinct cell death mechanisms, namely apoptosis, necroptosis, mitochondrial-mediated necrosis, pyroptosis, ferroptosis, and autophagic cell death (82). Excessive ROS levels and lipid peroxidation drive ferroptosis, a key mechanism of cell death in diabetic human heart failure, while matrix metalloproteinases (MMPs), particularly MMP9, promote fibrosis, thereby impairing contractility (83–86). This combination of cell death and fibrosis impairs myocardial contractility and increases stiffness, contributing to HFpEF and HFrEF (15, 87).

Emerging Mechanisms in Metabolic Remodeling of Diabetic Hearts

Metabolic remodeling in diabetic hearts is driven by complex molecular mechanisms that extend beyond traditional

metabolic pathways. These emerging mechanisms highlight the role of autophagy, epigenetic regulation, polyamine synthesis, and branched-chain amino acid (BCAA) metabolism in contributing to cardiac dysfunction. Understanding these processes may reveal novel therapeutic strategies to counteract the detrimental effects of diabetes on the heart.

Autophagy dysregulation.

Autophagy, the cellular process responsible for clearing damaged organelles and proteins, is disrupted in diabetes (88). Impaired mitophagy, a specific type of autophagy targeting damaged mitochondria, leads to the accumulation of dysfunctional mitochondria, further exacerbating oxidative stress and energy deficits (70, 89). The accumulation of damaged mitochondria increases ROS production, promoting cellular stress and cell death (90). Furthermore, impaired mitophagy, a mitochondria-selective form of autophagy that degrades and recycles damaged mitochondria, limits the heart's ability to adapt to metabolic stress, contributing to the energy deficits observed in DHF (70). Therapeutic strategies aimed at enhancing mitophagy and restoring proper autophagic function could offer potential interventions for reducing mitochondrial damage and improving cardiac function in diabetes.

Sirtuins and epigenetic regulation.

Sirtuins, particularly SIRT3, are mitochondrial deacetylases that regulate energy metabolism and oxidative stress (91). In diabetes, SIRT3 is downregulated, contributing to mitochondrial dysfunction and metabolic inflexibility (14). Epigenetic modifications also play a pivotal role in regulating gene expression in diabetic hearts (92). Histone acetylation and methylation influence pathways related to glucose and fatty acid metabolism, mitochondrial function, and oxidative stress responses (93, 94). These epigenetic changes can perpetuate maladaptive metabolic remodeling, further contributing to the progression of DHF. Targeting sirtuins and modulating epigenetic regulators may represent promising therapeutic strategies for reversing metabolic dysfunction in diabetic hearts.

Polyamines regulation.

Polyamines, small organic cations synthesized by cardiomyocytes, play a crucial role in maintaining cellular homeostasis and exhibit insulin-like activity (95, 96). In the context of diabetes, polyamines have been shown to enhance mitochondrial respiration, thereby improving cellular energy production and supporting metabolic remodeling within the diabetic heart (97). This enhancement in mitochondrial function is significant, as it helps mitigate the metabolic dysfunctions commonly associated with DMCM.

Branched-chain amino acid metabolism.

Disruption of branched-chain amino acid (BCAA) metabolism has emerged as a contributor to DHF (98). Accumulation of BCAA intermediates, due to reduced activity of branched-chain α -keto acid dehydrogenase, exacerbates metabolic stress, increases myocardial cell death, and impairs mitochondrial function. Targeting BCAA dysregulation may offer a novel therapeutic approach for mitigating DMCM. Additionally, polyamine regulation offers a potential therapeutic avenue, as enhanced mitochondrial respiration

through polyamine activity may help mitigate the energy deficits observed in diabetic hearts (96, 99). Furthermore, excess BCAAs have been linked to myocardial damage, increased cell death, and impaired cardiac function in experimental models, especially in T1DM (100). These findings suggest that targeting BCAA metabolism could offer a new avenue for therapeutic intervention in DHF, as it contributes to both metabolic and structural changes in the heart. Further research is needed to fully understand how BCAA dysregulation interacts with other metabolic pathways and its potential in developing treatments for heart failure in patients with diabetes.

CARDIAC SUBSTRATE METABOLISM IN DIABETES-INDUCED HEART FAILURE

DHF is a multifaceted condition driven by significant alterations in cardiac substrate metabolism. This metabolic remodeling, characterized by an increased reliance on fatty acids and a concomitant reduction in glucose utilization, leads to metabolic inflexibility, impaired cardiac efficiency, and ultimately heart failure (98, 101). The diabetic heart exhibits a unique metabolic phenotype, often described as a "fuel-starved" myocardium, where the availability and utilization of different substrates are compromised. Targeting these metabolic perturbations holds promise as a therapeutic strategy to mitigate the progression of heart failure in patients with diabetes (102). A study examining heart tissue from patients with diabetes and patients with non-diabetes during coronary artery bypass surgery found significant differences in mitochondrial function and oxidative stress. In patients with diabetes, there was a notable increase in the storage of triglycerides in the heart muscle, paired with a reduced ability of the mitochondria to use both carbohydrates (glutamate) and fats for energy production. Furthermore, patients with diabetes exhibited higher levels of hydrogen peroxide (H_2O_2) emissions from their mitochondria when oxidizing both carbohydrate and fat-based fuels, which is a sign of elevated oxidative stress. In addition, these patients had lower levels of glutathione, a crucial antioxidant, indicating depleted defenses against oxidative damage, and persistent oxidative stress in the heart tissue. This suggests that diabetes contributes to impaired energy metabolism and increased oxidative stress in the heart, potentially exacerbating cardiovascular risks (103). Emerging studies have expanded the understanding of the intricate interplay between metabolic pathways in the diabetic heart, suggesting that interventions to restore metabolic homeostasis could offer multifaceted benefits beyond conventional heart failure therapies (102). The following are the key metabolic substrates disrupted in the diabetic heart.

Fatty Acids

In diabetic hearts, there is a marked upregulation of fatty acid oxidation (FAO), driven by elevated circulating free fatty acids (FFAs) and increased expression of fatty acid transport proteins such as cluster of differentiation-36 (CD36) and fatty acid transporter protein (FATP) (104). Once inside the cell, fatty acids undergo β -oxidation in the mitochondria, a

process heavily dependent on carnitine palmitoyl transferases (CPT1 and CPT2). CPT1 facilitates the transport of long-chain fatty acids into the mitochondria by converting them into acyl-carnitine, whereas CPT2 enables their reconversion to acyl-CoA for subsequent β -oxidation. Dysregulation of CPT activity, often observed in heart failure, exacerbates the inefficient utilization of fatty acids, leading to increased lipid accumulation and mitochondrial dysfunction (105, 106). Although FAO is a critical energy-generating pathway for the heart, its overactivation in diabetes has several harmful effects, including increased oxygen consumption, lipid accumulation, and heightened oxidative stress (107). Fatty acids serve as a primary energy source, and their oxidation within mitochondria generates significant amounts of ATP. The ATP yield varies depending on the fatty acid's chain length and degree of saturation. For instance, palmitic acid (C16:0) produces 106 ATP while consuming 23 oxygen molecules, producing ~ 4.6 ATP per O_2 molecule. Similarly, stearic acid (C18:0) yields 120 ATP with a similar ATP/ O_2 ratio. Monounsaturated fatty acids like oleic acid (C18:1) and polyunsaturated fatty acids like linoleic acid (C18:2) produce slightly less ATP (118 and 116 ATP, respectively) due to the presence of double bonds, which reduces their energy efficiency. Arachidonic acid (C20:4), a polyunsaturated fatty acid, produces 144 ATP but requires 32 oxygen molecules, maintaining an ATP/ O_2 ratio of around 4.5. Overall, fatty acids are highly efficient in generating energy, with an average yield of 4.5–4.6 ATP per oxygen molecule, making them especially important during periods of high energy demand, such as prolonged exercise or fasting. However, in diabetic conditions, excessive reliance on FAO can disrupt the delicate balance of heart metabolism, exacerbating oxidative stress and lipid overload (44, 108).

To counteract these effects is the inhibition of key enzymes involved in FAO, such as carnitine palmitoyl transferase I (CPT1), the rate-limiting enzyme for mitochondrial fatty acid uptake. Pharmacological inhibitors of CPT1, such as etomoxir, have been shown to reduce fatty acid oxidation and improve glucose oxidation, thereby enhancing cardiac efficiency and function (109). However, clinical application of CPT1 inhibitors has been limited due to concerns about toxicity and adverse effects, particularly affecting the liver (110). Therefore, alternative strategies, such as partial inhibition of FAO or targeting downstream components of the FAO pathway, are being explored. For example, targeting peroxisome proliferator-activated receptor alpha (PPAR α), a key regulator of FAO, may offer a more balanced approach by modulating fatty acid metabolism without completely shutting down this vital pathway. Trimetazidine, an antianginal agent known for inhibiting 3-ketoacyl CoA thiolase, has been found to reduce fatty acid oxidation and alleviate diastolic dysfunction in mice subjected to a high-fat diet, a model for obesity-related cardiac impairment (111–113). Additionally, clinical studies have demonstrated that a 6-mo trimetazidine regimen improved both systolic and diastolic function in patients with T2DM (114). Trimetazidine's efficacy in improving myocardial efficiency without compromising overall cardiac energetics has positioned it as a promising candidate for broader clinical applications in diabetic heart disease. Recent trials continue to investigate its

long-term safety profile and potential synergistic effects when combined with other metabolic modulators (115). Efforts to reduce fatty acid uptake in cardiomyocytes have also yielded positive outcomes. Blocking the sarcolemmal fatty acid transporter FATP/CD36 with sulfo-*N*-succinimidyl oleate significantly reduced fatty acid metabolism in rodent models of T2DM (116). This intervention not only upregulated glycolysis but also improved cardiac function following ischemia.

Glucose

A hallmark of DMCM is the impaired utilization of glucose, primarily due to insulin resistance and downregulation of glucose transporters like GLUT4, the primary glucose transporter in the heart (117). Additional key mechanisms include reduced glycolytic enzymes hexokinase activity, the enzyme responsible for the phosphorylation of glucose to glucose-6-phosphate, a critical step in glucose metabolism; reduced activity of AMPK, the energy-sensing kinase, which diminishes the ability of cells to increase glucose uptake; and upregulation of transcription factor FOXO3 that promotes substrate utilization from glucose to fatty acids by altering the expression of enzymes and transporters involved in glucose metabolism (69, 118, 119). Restoring glucose utilization is a crucial therapeutic goal, as glucose oxidation is more oxygen-efficient than FAO, leading to improved ATP production and cardiac function (36, 120). This effect was linked to decreased dehydrogenase kinase 4 (Pdk4) expression and reduced dehydrogenase kinase phosphorylation, emphasizing the role of increased glucose oxidation in improving cardiovascular outcomes. Similar benefits were observed in T1DM models, where AS1842856 treatment improved cardiac function in streptozotocin-induced diabetic rats (121). Clinically, glucagon-like peptide 1 receptor (GLP-1R) agonists, such as liraglutide and semaglutide, have been linked to improved cardiovascular outcomes in patients with T2DM (122). These treatments have been associated with increased myocardial glucose oxidation and improved diastolic function in animal models and human studies, respectively. Studies also suggested combining GLP-1R agonists with sodium-glucose cotransporter 2 (SGLT2) inhibitors may provide additive benefits by enhancing glucose and ketone utilization pathways, thus optimizing cardiac metabolism in patients with diabetes (123–125).

Ketones

Recent research has highlighted the role of ketone bodies as an alternative fuel source in the diabetic heart (126). Ketone bodies, particularly β -hydroxybutyrate (β -HB), are produced in the liver from fatty acids and serve as an energy substrate when glucose availability is low, as observed in diabetes (77). Ketogenesis differs markedly between T1DM and T2DM due to variations in insulin levels and metabolic control. In T1DM, the absence of insulin leads to uncontrolled lipolysis, significantly increasing the availability of fatty acids for hepatic ketogenesis. This results in a substantial rise in circulating ketone bodies. In contrast, T2DM presents with a more moderate level of ketogenesis. This is due to partial insulin resistance and hyperinsulinemia, which partially suppresses lipolysis and, consequently, ketone body production.

The presence of residual insulin in T2DM helps prevent the excessive production of ketones observed in T1DM (127). This differential regulation impacts the metabolic profiles and cardiovascular risks in these conditions, with T1DM being more prone to ketoacidosis, whereas T2DM might experience a more chronic metabolic inflexibility (127).

Increased myocardial ketone oxidation has been proposed to enhance cardiac efficiency, particularly under conditions of reduced glucose availability. By serving as an alternative energy substrate, ketones support the heart's energy demands when glucose is limited (128). Moreover, ketones may function as signaling molecules, influencing key pathways involved in inflammation, oxidative stress, and cellular metabolism (129). This dual role in providing energy and regulating metabolic pathways underscores the potential of ketones to aid in the metabolic adaptation of the heart, especially in the context of diabetes or heart failure. Therapeutic strategies that modulate ketone body metabolism aim to optimize their utilization without triggering adverse effects. For instance, inhibitors of ketogenesis, such as the 3-hydroxy-3-methylglutaryl-CoA (HMG-CoA) synthase inhibitors, could be used to prevent excessive ketone production while allowing for their beneficial effects on the heart. Sodium-glucose cotransporter 2 (SGLT2) inhibitors, such as empagliflozin, have gained attention for their potential role in ketone metabolism. Although empagliflozin improved diastolic dysfunction in obese mice, it did not increase myocardial ketone oxidation in T2DM mice, suggesting that the cardiovascular benefits of SGLT2 inhibition may not be directly related to ketone metabolism and warrant further investigation (130). Recently, the EMPACT-MI trial found that empagliflozin significantly reduces the risk of first and total heart failure hospitalizations in patients with left ventricular dysfunction or congestion after acute myocardial infarction (131). Furthermore, ketone ester supplementation in T2DM mice improved diastolic and systolic function, likely due to enhanced mitochondrial respiration (132). However, the chronic utilization of ketone bodies presents potential risks, including the development of ketoacidosis and further metabolic inflexibility.

LIPOTOXICITY, MITOCHONDRIAL DYSFUNCTION, AND OXIDATIVE STRESS

Lipotoxicity, marked by the accumulation of toxic lipid species such as ceramides and diacylglycerol (DAG), occurs in both the cytoplasm and mitochondria (22, 133). Within mitochondria, these lipid intermediates impair oxidative phosphorylation and trigger mitochondrial permeability transition, leading to cell death. In the cytoplasm, lipid accumulation activates stress-related kinases (e.g., JNK, PKC) that perpetuate insulin resistance and inflammation. These distinct locations of lipid accumulation initiate separate but converging pathological cascades that exacerbate DHF (134, 135). Therapies aimed at reducing lipotoxicity are increasingly recognized for their potential to halt the progression of diabetic heart disease by restoring cellular homeostasis and preventing mitochondrial damage (136). Therapeutic strategies targeting lipotoxicity aim to reduce the accumulation of toxic lipids and restore mitochondrial function. One approach

is using agents that promote fatty acid oxidation in peripheral tissues, thereby reducing the availability of free fatty acids (FFAs) to the heart. Thiazolidinediones (TZDs), a class of peroxisome proliferator-activated receptor gamma (PPAR γ) agonists, have been shown to enhance adipose tissue lipid storage and reduce circulating FFAs, leading to improved cardiac metabolism. In addition, antioxidants that target mitochondrial ROS production, such as mitoTEMPO, have shown promise in reducing oxidative stress and preserving mitochondrial function in diabetic hearts (60). Recent reports demonstrated the combined use of TZDs and mitochondrial-targeted antioxidants to reduce lipid overload and oxidative damage in DMCM synergistically (137). Mitochondria-targeted antioxidants, such as MitoQ and coenzyme Q10, along with SIRT3 activators, are under investigation for their potential to restore mitochondrial function and mitigate oxidative damage (138). In addition, compounds that promote mitochondrial biogenesis, including resveratrol and PGC-1 α activators, have shown promise in enhancing myocardial energetics and improving cardiac function (139).

RECENT AND ONGOING CLINICAL TRIALS

Given the complexity of diabetes-induced metabolic alterations, a multimodal therapeutic approach may be necessary to target cardiac substrate metabolism and alleviate heart failure effectively (140). Such an approach could involve a combination of strategies, including partial FAO inhibition, enhancement of glucose and ketone utilization, and mitigation of lipotoxicity and oxidative stress. Evidence suggests that integrating multiple pharmacologic and lifestyle interventions tailored to individual metabolic profiles may yield the best outcomes for patients with diabetes, minimizing side effects while maximizing therapeutic benefits (141). Integrating these therapies could synergistically improve cardiac efficiency, reduce metabolic inflexibility, and ultimately improve outcomes in patients with diabetes with heart failure.

EMPA-REG OUTCOME Trial

Sodium-glucose cotransporter 2 (SGLT2) inhibitors, such as empagliflozin and dapagliflozin, have emerged as promising therapies for heart failure in patients with diabetes by promoting the use of ketones and reducing oxidative stress (142, 143). The Empagliflozin Cardiovascular Outcome Event Trial in Type 2 Diabetes Mellitus Patients—Removing Excess Glucose (EMPA-REG OUTCOME) trial showed that the SGLT2 inhibitor empagliflozin, when added to standard care, significantly reduced the risk of major cardiovascular events, cardiovascular deaths, hospitalizations for heart failure, and overall mortality in patients with T2DM who were at high cardiovascular risk (144). This was the first time a glucose-lowering drug has been shown to improve heart health outcomes in patients with T2DM.

DAPA-HF Trial

The Dapagliflozin and Prevention of Adverse Outcomes in Heart Failure (DAPA-HF) showed that dapagliflozin, an SGLT2 inhibitor, significantly reduces the risk of worsening heart failure or death from cardiovascular causes, regardless of the presence or absence of diabetes (145). This finding

underscores the potential of dapagliflozin not only as a glucose-lowering medication but also as a valuable treatment for heart failure in a broader patient population, offering important benefits in reducing heart failure progression and improving survival outcomes.

LIVE Trial

The LIVE trial (Effect of Liraglutide on Left Ventricular Function in Stable Chronic Heart Failure Patients) determined the effect of glucagon-like peptide-1 (GLP-1) analog liraglutide on the left ventricular function in patients with chronic heart failure with and without T2DM. The results showed that the GLP-1 agonist liraglutide did not improve left ventricular systolic function compared with placebo in patients with stable chronic heart failure, whether or not they had diabetes. Treatment with liraglutide was linked to an increased heart rate and more severe cardiac side effects, raising concerns about its use in patients with heart failure and reduced left ventricular function (146).

Other Trials

Pfizer conducted a 16-wk study to evaluate the efficacy and safety of PF-06882961 (danuglipron), a GLP-1 receptor agonist, in adults with T2DM. Danuglipron, which requires subcutaneous administration or fasting before oral use, was shown to significantly reduce HbA1c, fasting plasma glucose, and body weight compared with placebo by *week 16*. The drug's tolerability was consistent with its mechanism of action, demonstrating its potential for managing T2DM (147).

A recent study looked at how tirzepatide affects mortality and cardiovascular and kidney outcomes compared with GLP-1 receptor agonists (GLP-1 RAs) in people with T2DM. The study involved 14,834 patients treated with tirzepatide and 125,474 treated with GLP-1 RAs, all without advanced kidney disease or recent cardiovascular events. After a 10.5-mo follow-up, tirzepatide was linked to lower risks of death, heart problems, and kidney issues. It also helped lower blood sugar (HbA1c) and body weight more than GLP-1 RAs. These findings suggest that tirzepatide may provide greater benefits in reducing complications in patients with diabetes, supporting its use in treatment plans (148). Tirzepatide is approved by the FDA under the brand name **Mounjaro** for managing T2DM.

These clinical trials have highlighted the potential of various therapies to improve outcomes in patients with diabetic with heart failure by targeting metabolic pathways. The EMPA-REG OUTCOME trial demonstrated that the SGLT2 inhibitor empagliflozin significantly reduced cardiovascular events and mortality in patients with high-risk T2DM, marking a breakthrough for glucose-lowering drugs in heart health. Similarly, the DAPA-HF trial showed that dapagliflozin reduced heart failure progression and death risk in patients with both diabetes and nondiabetes. Conversely, the LIVE trial revealed that liraglutide, a GLP-1 agonist, did not improve heart function in patients with chronic heart failure and was linked to adverse cardiac effects. Further research on GLP-1 receptor agonists, such as danuglipron and tirzepatide, showed positive results in lowering HbA1c, weight, and cardiovascular risks, with tirzepatide outperforming GLP-1 RAs in reducing complications,

supporting its use in managing T2DM. These trials underscore the importance of tailored, multimodal treatment strategies for patients with diabetes and heart failure.

METHODS TO ASSESS SUBSTRATE METABOLISM AND MYOCARDIAL ENERGETICS

Various sophisticated methods are used to assess these parameters, each offering unique insights and advantages. This section discusses the key techniques used in contemporary research to elucidate the metabolic and energetic alterations in DHF.

Metabolomics

Metabolomics comprehensively analyzes small molecules within biological samples, offering crucial insights into the metabolic alterations associated with DHF (HF) (149). This approach involves three major methods: nuclear magnetic resonance (NMR) spectroscopy, mass spectrometry (MS)-based techniques, and stable isotope-resolved metabolomics (SIRM) with ^{13}C metabolic flux analysis.

Nuclear magnetic resonance spectroscopy metabolomics.

Nuclear magnetic resonance (NMR) exploits the spin property of specific atomic nuclei to visualize chemical structures. In the presence of a magnetic field, these nuclei transition between different energy states, which can be detected and analyzed through radio frequency signals. NMR spectroscopy is a powerful tool for metabolic profiling due to its nondestructive nature and quantitative accuracy. It enables the identification and quantification of metabolites without extensive sample preparation. In the context of diabetes-induced DHF, NMR spectroscopy has uncovered metabolic remodeling in cardiac tissues (150). For example, in a study by Zhang et al. (151), proton (^1H) NMR spectroscopy was used to profile myocardial metabolites in diabetic rats, revealing significant alterations in glucose and lipid metabolism pathways. This ability to provide detailed metabolic fingerprints makes it indispensable for studying the metabolic derangements in diabetic hearts.

Mass spectrometry-based metabolomics.

Mass spectrometry begins with the ionization of analytes in the sample, followed by the separation and detection of these ions based on their mass-to-charge (m/z) ratio (152). Mass spectrometry-based metabolomics, including liquid chromatography-mass spectrometry (LC-MS) and gas chromatography-mass spectrometry (GC-MS), offers high sensitivity and specificity for detecting a wide range of metabolites. LC-MS has been widely used to analyze alterations in fatty acid metabolism in DMCM. A study by Randle et al. used LC-MS to demonstrate an accumulation of acylcarnitines and a reduction in ketone body utilization in the human hearts of diabetic mice. GC-MS, on the other hand, has been instrumental in profiling volatile organic compounds and intermediates of the tricarboxylic acid (TCA) cycle, providing insights into the metabolic flux alterations in diabetic hearts.

Stable isotope-resolved metabolomics and ^{13}C metabolic flux analysis.

Stable isotope-resolved metabolomics (SIRM) combined with ^{13}C metabolic flux analysis offers dynamic insights into metabolic pathways by tracing labeled substrates through metabolic networks. This approach has been instrumental in elucidating the flux of glucose and fatty acids in diabetic hearts (153). Multiple techniques are available for delivering stable-isotope tracers in vivo or ex vivo, such as oral administration, intravenous infusion, and heart perfusion. Ex vivo perfusion techniques, like the Langendorff method and working heart preparation, are essential for studying cardiac metabolism, with the latter providing more physiological contractile function and preventing anoxia. Such studies underscore the utility of SIRM in mapping metabolic fluxes and identifying key regulatory nodes altered in diabetes-induced HF.

Positron emission tomography.

Positron emission tomography (PET) provides noninvasive, quantitative imaging of metabolic processes in vivo. In DHF, PET has been pivotal in assessing myocardial glucose and fatty acid uptake. A study by Shao et al. (154) used 18F-fluorodeoxyglucose (FDG) PET to demonstrate reduced myocardial glucose uptake in diabetic mice, correlating with disease severity. Similarly, a study by Welch et al. demonstrated that small-animal PET, using tracers such as ^{15}O -water, ^{11}C -glucose, ^{11}C -acetate, and ^{11}C -palmitate, can noninvasively quantify alterations in myocardial perfusion and substrate metabolism in diabetic rats. The study found significantly higher myocardial glucose and fatty acid utilization and oxidation in Zucker diabetic fatty rats compared with their lean littermates (155). The ability of PET to provide metabolic imaging makes it a valuable tool for studying cardiac metabolism in diabetic conditions.

^{31}P nuclear magnetic resonance spectroscopy and ^{31}P magnetic resonance spectroscopy.

^{31}P NMR spectroscopy and magnetic resonance spectroscopy (MRS) offer insights into myocardial energetics by quantifying high-energy phosphate compounds such as ATP and phosphocreatine (PCr). These methods have revealed crucial information about the energetic deficits in diabetic hearts. For example, Neubauer et al. (156) used ^{31}P MRS to show decreased PCr/ATP ratios in patients with diabetes with HF, indicating compromised myocardial energy reserve. A study by Valković et al. demonstrated that phosphorus magnetic resonance spectroscopy (^{31}P -MRS) at 7 T revealed an increased cardiac Pi/PCr ratio in patients with type 2 diabetes compared with healthy controls. This finding indicates altered cardiac metabolism in diabetes, although no significant correlation was found between Pi/PCr and PCr/ATP (157). This reduction in energy reserve correlates with impaired cardiac function, emphasizing the importance of energetic assessment in understanding DMCM.

Hyperpolarized magnetic resonance imaging.

Hyperpolarized magnetic resonance imaging (HP-MRI) enhances the sensitivity of MR techniques by hyperpolarizing nuclei, significantly increasing signal strength. This method

has been used to study metabolic intermediates in real time. In DHF, HP-MRI with hyperpolarized [$1\text{-}^{13}\text{C}$] pyruvate has revealed reduced pyruvate dehydrogenase (PDH) activity and altered lactate production, indicative of impaired glucose oxidation. HP-MRI thus provides dynamic insights into metabolic pathway activities and their alterations in diabetic hearts. A study by Rider et al. demonstrated the use of hyperpolarized ^{13}C MRI to assess cardiac metabolism in individuals with T2DM compared with healthy controls. The study highlighted reduced metabolic flux through pyruvate dehydrogenase in T2DM and its potential to detect and quantify metabolic changes in cardiovascular metabolism in diabetic hearts (158).

Mitochondrial respiration assays.

Mitochondrial respiration is a critical indicator of cellular bioenergetic health, typically measured using intact myocardial tissue, permeabilized cardiac fibers, or cell-based models. High-resolution respirometry systems, such as Oroboros or Seahorse XF, allow for precise assessment of mitochondrial function by measuring the oxygen consumption rate (OCR) under various experimental conditions. These systems enable the evaluation of mitochondrial respiration applying different substrates, such as fatty acids, glutamate, or malate, and chemical inhibitors targeting specific mitochondrial complexes, like rotenone or oligomycin. Intact myocardial tissue provides the advantage of assessing mitochondrial function within its natural cellular environment, whereas permeabilized cardiac fibers allow direct access to mitochondria, making it a preferred method for investigating mitochondrial bioenergetics in disease models like DMCM. Cell-based models, on the other hand, offer the ability to manipulate mitochondrial pathways genetically or pharmacologically, enabling a more controlled exploration of mitochondrial function. This approach enables a comprehensive analysis of mitochondrial respiration, including basal respiration, ATP-linked respiration, proton leak, and maximal respiratory capacity. Such detailed assessments are invaluable in understanding mitochondrial dysfunction in cardiovascular diseases, diabetes, and metabolic disorders, providing critical insights into potential therapeutic targets aimed to restore mitochondrial health. High-resolution respirometry has been used to measure oxygen consumption rates in isolated mitochondria from diabetic hearts (159). A study by Guarini et al. demonstrated that mitochondrial respiration is impaired in obesity and diabetes, leading to mitochondrial DNA damage, and reduced coronary metabolic dilation. Repairing mtDNA damage in obese fatty rats restored mitochondrial function and improved metabolic dilation, highlighting the critical role of mitochondrial integrity in linking myocardial blood flow to metabolism (160). These assays highlight the significant role of mitochondrial energetics in maintaining cardiac function and the impact of diabetes on mitochondrial health.

Gene expression profiling.

Gene expression profiling using techniques such as RNA sequencing (RNA-seq) and microarrays enables the identification of dysregulated metabolic pathways at the transcriptional level. In DMCM, gene expression studies have revealed altered expression of genes involved in glucose and

lipid metabolism, mitochondrial function, and oxidative stress (161). For example, Kambis et al. used RNA-seq to identify the downregulation of genes encoding key glycolytic enzymes and the upregulation of those involved in fatty acid oxidation in diabetic mouse hearts. These findings provide a molecular basis for the metabolic remodeling observed in diabetic hearts.

Proteomics.

Proteomics, the large-scale study of proteins, offers insights into the functional consequences of metabolic alterations in DMCM. Mass spectrometry-based proteomics has identified changes in the abundance and post-translational modifications of proteins involved in energy metabolism, contractile function, and oxidative stress. For instance, a study by Hamblin et al. (162) use quantitative proteomics to reveal reduced levels of key enzymes in the TCA cycle and electron transport chain in diabetic rat hearts, correlating with impaired mitochondrial function. Proteomic analyses thus complement metabolomic and genomic studies, providing a holistic view of the metabolic dysregulation in DHF.

In summary, the comprehensive assessment of substrate metabolism and myocardial energetics is crucial for advancing our understanding of DHF. These methodologies provide detailed insights into the metabolic and energetic alterations underlying the disease, enabling the identification of potential therapeutic targets. By integrating data from metabolomics, imaging, mitochondrial function assays, gene expression profiling, and proteomics, researchers can construct a detailed map of the metabolic dysfunction in DHF. This integrated approach facilitates the development of targeted interventions aimed at restoring metabolic homeostasis and improving cardiac function in patients with diabetic.

EMERGING THERAPIES AND FUTURE DIRECTIONS

The growing understanding of diabetes-induced cardiac metabolism offers promising research avenues. Here are some key directions:

Targeting Metabolic Pathways

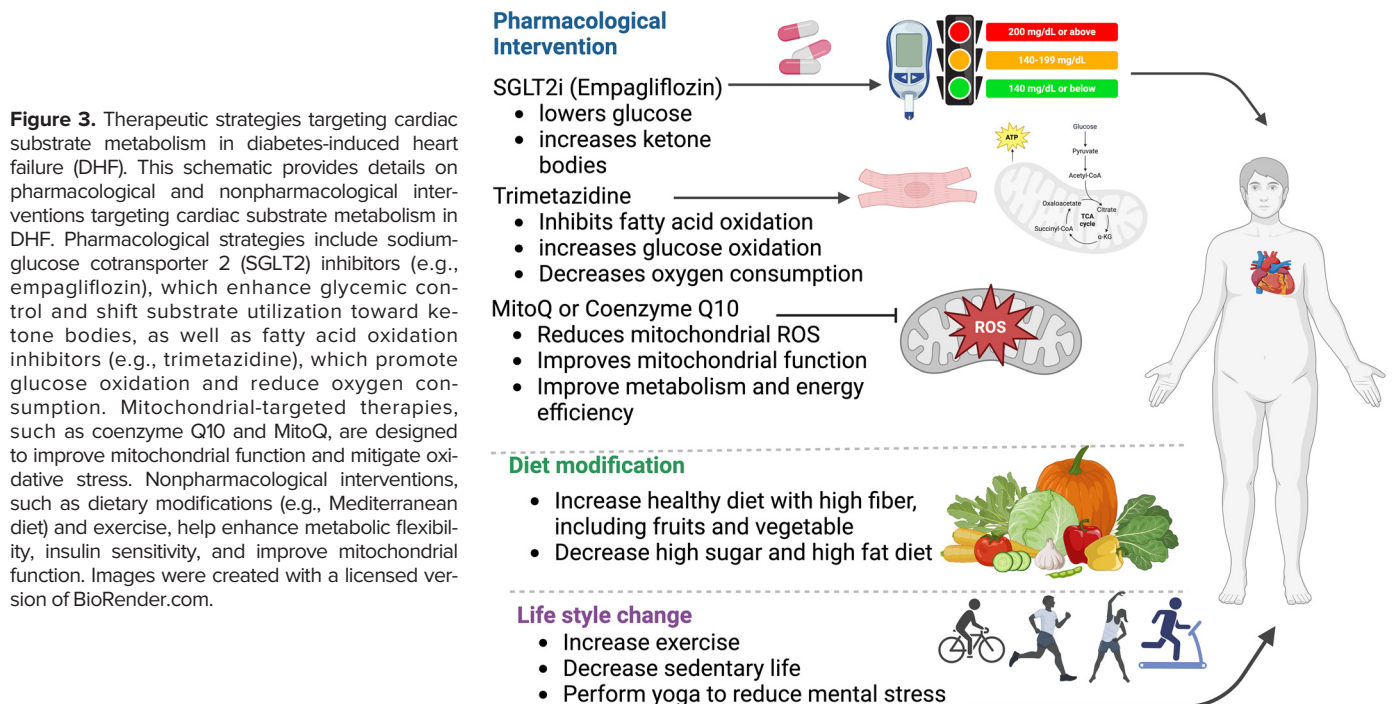
One of the primary focuses is identifying and targeting disrupted metabolic pathways in DMCM, such as abnormal fatty acid oxidation and impaired glucose utilization. Future research could explore therapies aimed at restoring the balance between these pathways, improving energy efficiency, and reducing the accumulation of toxic metabolites like ceramides and lipid intermediates, which are associated with cardiac dysfunction.

Mitochondrial Function and Biogenesis

Mitochondrial dysfunction plays a vital role in DHF. Future investigations should explore ways to enhance mitochondrial biogenesis, optimize ATP production, and improve the mitochondrial quality control process by mitophagy. Mitochondrial-targeted therapies may potentially prevent or reverse the energetic deficits that occur in diabetic hearts.

Integration of Exercise and Diet with Pharmacotherapy

Exploring the synergistic effects of lifestyle interventions, such as exercise and dietary modifications, in combination with novel pharmacotherapies (e.g., SGLT2 inhibitors or GLP-1 receptor agonists) may lead to more comprehensive treatment approaches. Understanding how these interventions interact on a molecular level in the diabetic heart can further optimize heart failure management strategies.



Adopting a heart-healthy diet, such as the Mediterranean diet, has improved metabolic parameters and reduced cardiovascular risk in patients with diabetes. This diet emphasizes the consumption of whole grains, fruits, vegetables, lean proteins, and healthy fats, which collectively improve glycemic control, reduce oxidative stress, and enhance cardiac function (163). Also, calorie restriction and intermittent fasting have been explored as strategies to improve metabolic flexibility and mitochondrial function, with studies showing improved insulin sensitivity and reduced oxidative stress in individuals with diabetes (164). Regular physical activity is a cornerstone of diabetes management and significantly benefits cardiac health. Aerobic exercise improves glucose uptake, enhances insulin sensitivity, and promotes mitochondrial biogenesis, thereby improving myocardial energetics and reducing the risk of heart failure (165). Resistance training also has beneficial effects on muscle metabolism and cardiac function. Studies have indicated that exercise training improves left ventricular function and reduces cardiovascular events in patients with diabetes (166).

In addition, bariatric surgery has helped treat patients with a body mass index (BMI) > 40 or with a BMI > 35 and the presence of a major obesity-related comorbidity. However, performance of a bariatric procedure appears to have marked benefits in patients with diabetes with a BMI even less than 30 (167, 168). In patients with T1DM diabetes and patients with diabetes with renal failure, the pancreas transplant is very effective at controlling diabetes (169, 170).

SUMMARY AND CONCLUSIONS

DHF presents a persistent clinical challenge, driven by intricate metabolic remodeling that encompasses insulin resistance, mitochondrial dysfunction, and oxidative stress. Unraveling the molecular pathways behind these alterations, along with using advanced imaging techniques to study them, opens doors for the development of targeted therapies. Modulating cardiac substrate metabolism through both pharmacological and lifestyle interventions holds promise as an effective strategy for managing diabetes-related heart failure. By enhancing metabolic flexibility, improving mitochondrial function, and optimizing substrate utilization, these approaches have the potential to alleviate the burden of heart failure in patients with diabetes significantly (Fig. 3). Continued research into the mechanisms of cardiometabolic remodeling will further illuminate novel therapeutic strategies.

In conclusion, targeting cardiac substrate metabolism represents a promising avenue for treating DHF. Although considerable progress has been made in understanding the underlying metabolic derangements, further research is needed to refine these therapeutic strategies and facilitate their translation into clinical practice. Developing safe and effective agents that can modulate multiple aspects of cardiac metabolism holds the potential for improving the prognosis for patients with diabetes and heart failure.

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DISCLOSURES

No conflicts of interest, financial or otherwise, are declared by the authors.

AUTHOR CONTRIBUTIONS

G.S. and P.K.M. conceived and designed research; P.K.M. prepared figures; G.S., S.S.C., M.A.C., and P.K.M. drafted manuscript; G.S., S.S.C., M.A.C., and P.K.M. edited and revised manuscript; G.S., S.S.C., M.A.C., and P.K.M. approved final version of manuscript.

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